

HW 1

Problem 1

1.

$$\{(1, 1), (1, -1)\}$$

2.

$$\{(1, -1, 1), (1, -1, -1), (-1, 1, 1), (-1, 1, -1)\}$$

3.

$$A_n = \{(a_1, a_2, \dots, a_n, \dots) \in \Omega \mid \sum_{i=1}^n a_i = 0\}$$

4.

$$\bigcup_{n=1}^{\infty} A_n$$

5.

No for example the infinite sequence of +1's

Problem 2

1.

Since $\Omega = \bigcup_{n \in N} \{n\}$ by definition

$$\begin{aligned}\mathbb{P}(\Omega) &= \sum_{n \in N} \mathbb{P}(\{n\}) = 1 \\ \sum_{n \in N} C_s n^{-s} &= 1 \\ C_s \sum_{n \in N} n^{-s} &= 1 \\ C_s &= \frac{1}{\sum_{n \in N} n^{-s}}\end{aligned}$$

2.

Since $A = \{2, 4, 6, \dots\} = \bigcup_{n \in N} \{2n\}$ by definition

$$\begin{aligned}\mathbb{P}(A) &= \sum_{n \in N} \mathbb{P}(\{2n\}) \\ &= \sum_{n \in N} C_s (2n)^{-s} \\ &= C_s 2^{-s} \sum_{n \in N} n^{-s}\end{aligned}$$

From 1. we know $C_s = \frac{1}{\sum_{n \in N} n^{-s}}$

$$= \frac{1}{2^s}$$

3.

Since $A = \{q, 2q, 3q, 4q, \dots\} = \bigcup_{n \in N} \{qn\}$ by definition

$$\begin{aligned}\mathbb{P}(A) &= \sum_{n \in N} \mathbb{P}(\{pn\}) \\ &= \sum_{n \in N} C_s (pn)^{-s} \\ &= C_s p^{-s} \sum_{n \in N} n^{-s}\end{aligned}$$

From 1. we know $C_s = \frac{1}{\sum_{n \in N} n^{-s}}$

$$= \frac{1}{p^s}$$

Do the same thing for B and we get $\mathbb{P}(B) = \frac{1}{q^s}$

Do the same thing for $A \cap B = \{pq, 2pq, 3pq, \dots\}$ and we get $\mathbb{P}(A \cap B) = \frac{1}{(pq)^s}$

$$\mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A \cap B) = \frac{1}{(pq)^s}$$

Thus by definition event A and event B are independent with respect to this measure $\mathbb{P}$

Problem 3**1.**

First consider $A \cap B = (A^c \cup B^c)^c$

$$\begin{aligned}\mathbb{P}(A \cap B) &= \mathbb{P}((A^c \cup B^c)^c) \\ &= 1 - \mathbb{P}(A^c \cup B^c)\end{aligned}$$

Proven in class $\mathbb{P}(A^c \cup B^c) \leq \mathbb{P}(A^c) + \mathbb{P}(B^c)$

$$\begin{aligned}&\geq 1 - \mathbb{P}(A^c) - \mathbb{P}(B^c) \\ &= 1 - 1 + \mathbb{P}(A) - 1 + \mathbb{P}(B) \\ &= \mathbb{P}(A) + \mathbb{P}(B) - 1\end{aligned}$$

Thus $\mathbb{P}(A \cap B) \geq \mathbb{P}(A) + \mathbb{P}(B) - 1$

Now consider $(A \cap B) \cap C$ apply the proof above to these sets.

$$\mathbb{P}((A \cap B) \cap C) \geq \mathbb{P}(A \cap B) + \mathbb{P}(C) - 1$$

Now apply proof again

$$\geq \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - 2$$

Therefore $\mathbb{P}((A \cap B) \cap C) \geq \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - 2$

2.